

NACS 645 – Model-free vs Model-based

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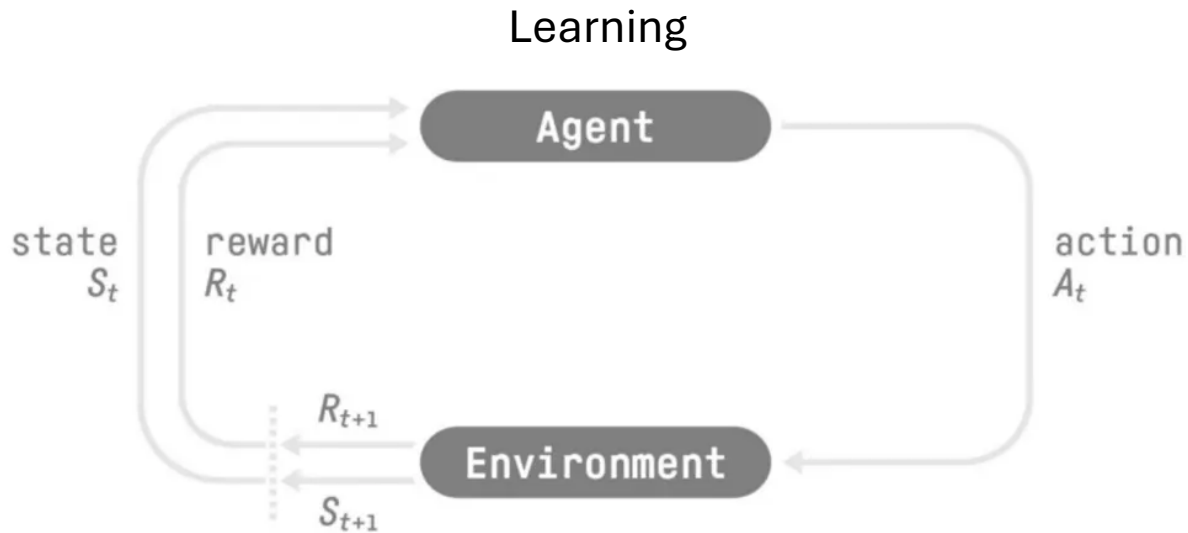


DEPARTMENT OF
PSYCHOLOGY



PROGRAM IN
NEUROSCIENCE &
COGNITIVE SCIENCE

RL from the algorithms perspective



All adaptation arises solely from the sequence of states, actions, and rewards generated by interaction with the environment.

Reinforcement learning is a framework for solving decision problems.

Agents learn from the environment by interacting with it through trial and error.

They receive rewards/punishments as feedback (no supervision).

Repeatedly:

- The agent receives **state** S_0 from the **Environment**
- Based on that **state** S_0 , the Agent takes **action** A_0
- The environment goes to a **new state** S_1
- The environment gives some **reward** R_1 to the Agent

Through this trial-and-error cycle, the agent adjusts its behaviour to **maximize expected cumulative reward** (reward hypothesis).

Markov Decision processes

A Markov Decision Process (MDP) is the standard formalism for reinforcement learning.

It defines an environment in terms of:



The **Markov property** states that optimal decisions depend only on the *current* state, not on the full history.

The agent's objective is to select actions that maximize the **discounted cumulative reward**, weighting future rewards less than immediate ones.

- **Observation/States:** information describing the current situation
 - *Fully observed* setting: a **state** provides a **complete description of the world** (chess)
 - *Partially observed* setting: an **observation** provides a **partial description of the state** (poker)
- **Actions:** the set of all moves the agent can take
 - Discrete: the number of possible actions is **finite**
 - Continuous: the number of possible actions is **infinite**
- **Rewards:** scalar indicating the immediate consequence of an action

Rewards and discounting

Cumulative reward at each time step t

$$R(\tau) = r_{t+1} + r_{t+2} + r_{t+3} + r_{t+4} + \dots$$



Return: cumulative reward

Trajectory (read Tau)
Sequence of states and actions

$$R(\tau) = \sum_{k=0}^{\infty} r_{t+k+1}$$

Discounted expected cumulative reward
at each time step t

$$R(\tau) = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \gamma^3 r_{t+4} + \dots$$



Return: cumulative reward

Gamma: discount rate

Trajectory (read Tau)
Sequence of states and actions

$$R(\tau) = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$

If the agent only exploits that they learnt so far, they may neglect alternatives with higher payoff, and not maximize their expected reward.

Hence, there is an **exploration/exploitation tradeoff** the agent should solve.

Value-based methods

Policy-Based Methods

- Learn the **policy directly** (*state* -> *action* mapping)
 - The model outputs the **action** (or a probability distribution over actions) **for each state**
 - The agent improves this mapping through experience
- The agent “learns how to act” without explicitly evaluating states.

Value-Based Methods

- Learn a **value function** (*state* -> *expected return* mapping)
- $V(s)$ or $Q(s,a)$ estimates “how good” it is to be in a state or to take an action
- The policy is **derived** by choosing actions that lead to higher estimated values

The agent “learns which states are better” and acts by following a policy (i.e., moving toward higher-value states).

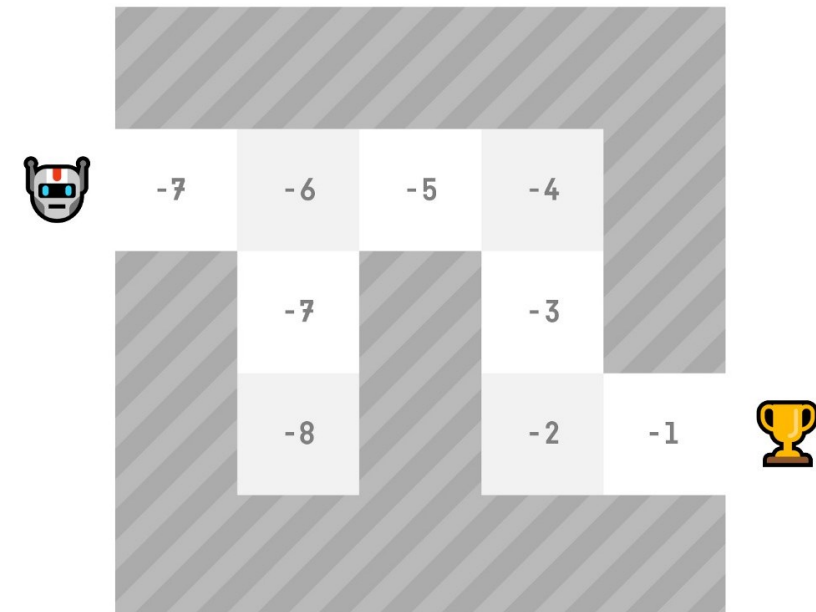
Value-based

$$\underline{v_{\pi}(s)} = \mathbb{E}_{\pi} [\underline{R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots} \mid \underline{S_t = s}]$$

Value
function

Expected discounted return

Starting
at state s



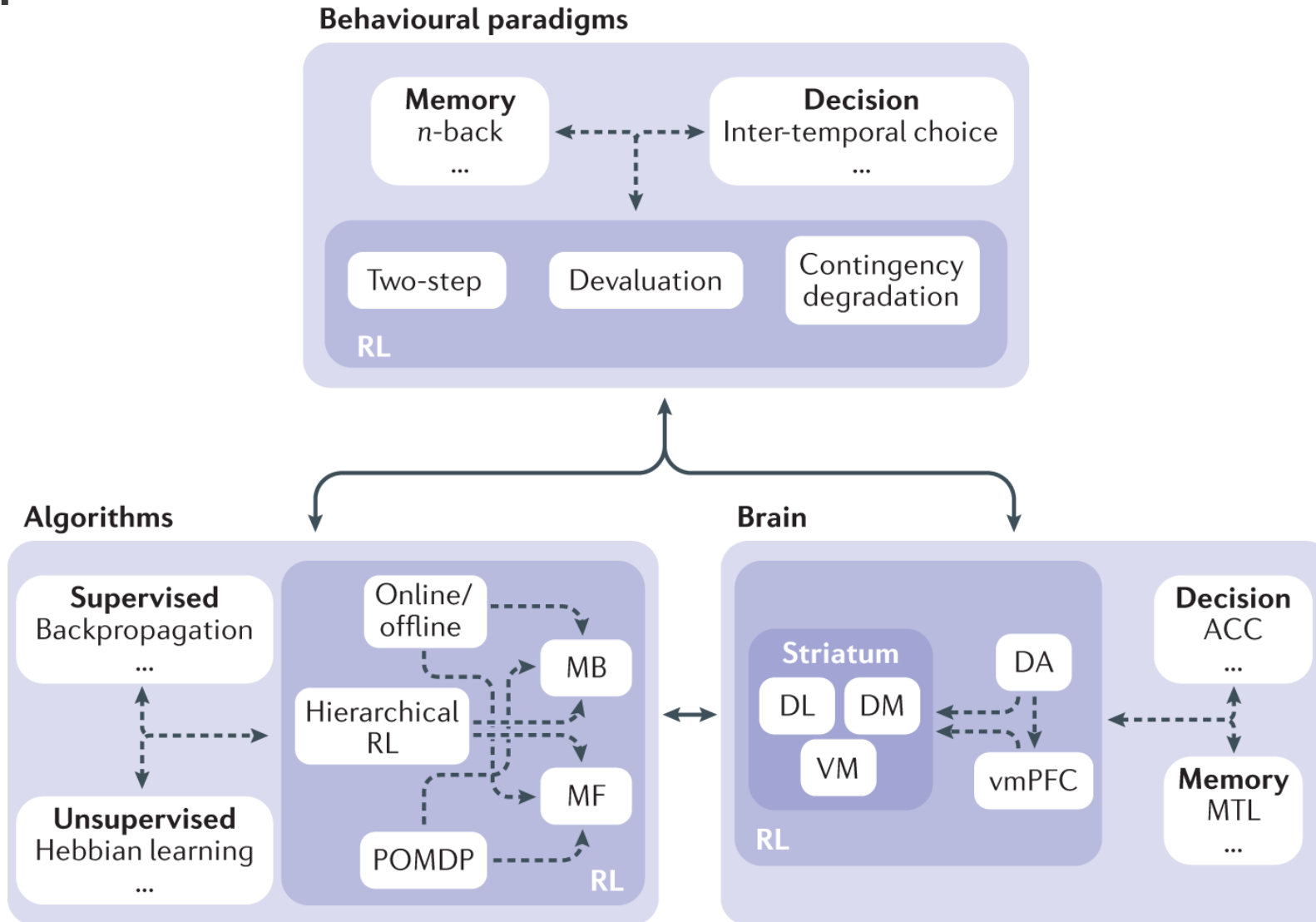
RL across fields of research

RL is a formal language for learning-based, sequential decision-making. It gives tools to talk about goals, prediction, uncertainty, credit assignment.

Early behavior studies (1890s–1950s):
Trial-and-error learning, conditioning, habit
vs goal-directed actions

Algorithmic RL (1950s–1990s): Formal mathematical tools for sequential decision-making: Dynamic programming, MDPs, TD learning, Q-learning

Neural interpretations (1990s–2000s):
Dopamine prediction-error signals; striatal habit learning; prefrontal planning.
Computational RL imported into neuroscience



Glossary

- **Reinforcement Learning (RL)**
Framework in which an agent learns through interaction with an environment to maximize expected cumulative reward.
- **Markov Decision Process (MDP)**
Formal model defining states, actions, transition dynamics, and rewards under the Markov property (current state contains all decision-relevant information).
- **State**
Complete description of the environment at a time point. In a fully observed setting, no hidden variables remain.
- **Observation**
Partial information available to the agent when the environment is not fully observed.
- **Action**
A choice available to the agent. Action spaces may be discrete or continuous.
- **Transition Function**
Specification of how states evolve when actions are taken: $P(s' | s, a)$.
- **Reward Function**
Scalar feedback indicating the immediate consequence of an action in a state.
- **Discounted Return**
Sum of future rewards, discounted to give more weight to near-term outcomes.
- **Policy (π)**
A decision rule mapping states to actions, deterministic or stochastic.
- **Policy-Based RL**
Learn the policy directly (state $\rightarrow$ action). Optimization focuses on improving behaviour.
- **Value-Based RL**
Learn a value function (state $\rightarrow$ expected return, or state-action $\rightarrow$ expected return). The policy is derived from these values.
- **Value Function (V, Q)**
 $V(s)$: expected return from state s .
 $Q(s,a)$: expected return from taking action a in state s and following the policy.
- **Model-Free RL (MF)**
Learns value estimates or policies directly from experience without an explicit transition model.
- **Model-Based RL (MB)**
Learns or uses a model of transitions and rewards to evaluate future outcomes via planning.
- **Markov Property**
The current state contains all the information needed for optimal decision-making; the past is irrelevant given the present.